



Yakeen Plus (2025)

PRACTICE TEST - 03

DURATION : 200 Minutes

DATE : 29/09/2024

M. MARKS : 720

Topics Covered

Physics :	Work energy and power, Centre of mass and System of Particles, Rotational Motion
Chemistry :	Thermodynamics & Thermochemistry, Chemical Equilibrium Ionic Equilibrium
Biology :	(Botany): The Living World (Zoology): Body Fluids and Circulation

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I (**Physics**), Section-II (**Chemistry**), Section-III (**Botany**) & Section IV (**Zoology**) and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

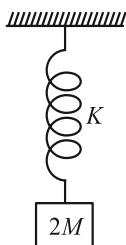
OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

SECTION-(I) PHYSICS

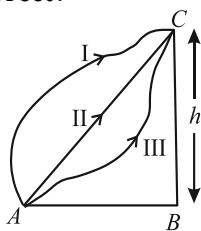
SECTION - A

1. A block of mass $2M$ is held at rest when the spring is at its natural length as shown. Now the block is released suddenly in this position to fall vertically. Find the maximum elongation in the spring. (Spring is ideal)



- (1) $\frac{2Mg}{K}$ (2) $\frac{4Mg}{K}$
 (3) $\frac{Mg}{K}$ (4) $\frac{Mg}{2K}$

2. A particle is to be moved from point A to C via paths (I), (II) and (III) in conservative force field as shown in figure, then which of the following options is **correct**?



- (1) Work done by conservative force is same for all the path.
 (2) Work done by conservative force is minimum for path (II).
 (3) Work done by conservative force is maximum for path (I).
 (4) None of the above.

3. Given below are two statements:
 If two springs S_1 and S_2 of force constants k_1 and k_2 , respectively, are stretched by the same force, it is found that more work is done on spring S_1 than on spring S_2 .

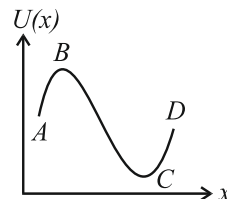
Statement I: If stretched by the same amount, work done on S_1 , will be more than that on S_2 .

Statement II: $k_1 < k_2$.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

4. The potential energy of a particle varies with distance x as shown in the graph. The force acting on the particle is zero at.



- (1) C only
 (2) B only
 (3) B and C
 (4) A and D

5. A force $F = (3x^2 + 2x - 7)$ N acts on a 2 kg body as a result of which the body gets displaced from $x = 0$ to $x = 5$ m. The work done by the force will be:

- (1) 35 J
 (2) 70 J
 (3) 115 J
 (4) 270 J

6. A body of mass 6 kg under a force which caused displacement in it given $S = \frac{t^2}{4}$, metres where ' t ' is time. The work done by the force in 2 seconds is:

- (1) 12 J
 (2) 9 J
 (3) 6 J
 (4) 3 J

7. Match List-I with List-II:

List-I		List-II	
(A)	Gravitational force	(I)	Decreases
(B)	Frictional force	(II)	Conservative force
(C)	KE of a dropped body	(III)	Non-conservative force
(D)	PE of a dropped body	(IV)	Increases

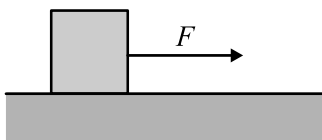
Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
 (2) A-III, B-I, C-IV, D-II
 (3) A-I, B-III, C-IV, D-II
 (4) A-II, B-IV, C-III, D-I

8. A cord is used to lower vertically a block of mass M by a distance d with constant downward acceleration $\frac{g}{4}$. Work done by the cord on the block is:

- (1) $Mg\frac{d}{4}$
- (2) $3Mg\frac{d}{4}$
- (3) $-3Mg\frac{d}{4}$
- (4) $-Mgd$

9. A 10 kg block placed on a rough horizontal floor is being pulled by a constant force of 50 N. Coefficient of kinetic friction between the block and the floor is 0.4. The total work done by all forces acting on the block over a displacement of 5 m is: ($g = 10 \text{ m/s}^2$)



- (1) 50 J
 - (2) 250 J
 - (3) 200 J
 - (4) 450 J
10. A force of $(2\hat{i} + 3\hat{j} + 4\hat{k})$ N acts on a body for 4 second, produces a displacement of $(3\hat{i} + 4\hat{j} + 5\hat{k})$ m. The power used is:
- (1) 9.5 W
 - (2) 7.5 W
 - (3) 6.5 W
 - (4) 4.5 W
11. A spring gun of spring constant 90 N/cm is compressed 12 cm by a ball of mass 16 g. If the trigger is pulled, the velocity of the ball is:
- (1) 90 ms^{-1}
 - (2) 9 ms^{-1}
 - (3) 40 ms^{-1}
 - (4) 60 ms^{-1}
12. Which of the following forces is called a conservative force?
- (1) Frictional force
 - (2) Air resistance
 - (3) Electrostatic force
 - (4) Viscous force

13. The potential energy of a particle in a force field is $U = \frac{A}{r^2} - \frac{B}{r}$ where A and B are positive constants and r is the distance of the particle from the centre of the field. For stable equilibrium, the distance of the particle is:

- (1) $\frac{B}{2A}$
- (2) $\frac{2A}{B}$
- (3) $\frac{A}{B}$
- (4) $\frac{B}{A}$

14. A particle moves from a point $(-2\hat{i} + 5\hat{j})$ m to $(4\hat{j} + 3\hat{k})$ m when a force of $(4\hat{i} + 3\hat{j})$ N is applied. How much work has been done by the force?

- (1) 8 J
- (2) 11 J
- (3) 5 J
- (4) 2 J

15. When a long spring is stretched by 2 cm, its potential energy is U . If the spring is stretched by 10 cm, the potential energy stored in it will be:

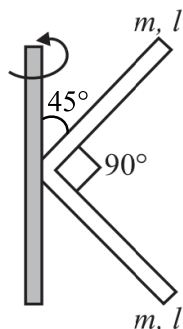
- (1) $U/5$
- (2) $5U$
- (3) $10U$
- (4) $25U$

16. The position vector of three particles of masses $m_1 = 1 \text{ kg}$, $m_2 = 2 \text{ kg}$ and $m_3 = 4 \text{ kg}$ are $\vec{r}_1 = (\hat{i} + \hat{j})$ m, $\vec{r}_2 = (2\hat{i} + 5\hat{j})$ m and $\vec{r}_3 = (2\hat{i} - \hat{j})$ m respectively. The position vector of their centre of mass is:

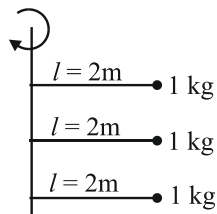
- (1) $(7\hat{i} - 13\hat{j})$ m
- (2) $\left(\frac{13\hat{i} + 17\hat{j}}{7}\right)$ m
- (3) $\left(\frac{13\hat{i} + 7\hat{j}}{7}\right)$ m
- (4) $(13\hat{i} + 17\hat{j})$ m

17. The position of centre of mass of system of particles at any moment does not depend on:
- (1) masses of the particles.
 - (2) forces on the particles.
 - (3) positions of the particles.
 - (4) relative distances between the particles.

18. Two identical uniform rods each of mass m and length l are joined perpendicular to each other. An axis passes through junction and in the plane of rods. Then M.I. of system about the axis is:



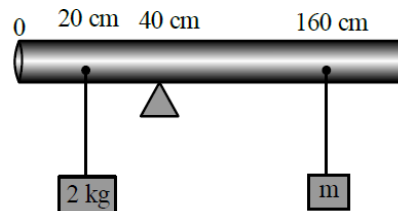
- (1) $\frac{1}{3}ml^2$
 - (2) $\frac{1}{3\sqrt{2}}ml^2$
 - (3) ml^2
 - (4) $\frac{ml^2}{2}$
19. Find out moment of inertia of point masses about given axis as shown:



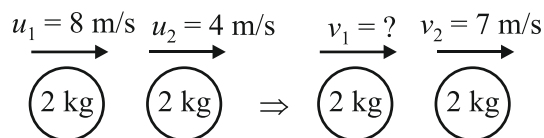
- (1) 4 kg.m^2
 - (2) 8 kg.m^2
 - (3) 12 kg.m^2
 - (4) 16 kg.m^2
20. A thin rod of mass M and length L is bent into a circular ring. The expression for moment of inertia of ring about an axis passing through its diameter is:

- (1) $\frac{ML^2}{2\pi^2}$
- (2) $\frac{ML^2}{4\pi^2}$
- (3) $\frac{ML^2}{8\pi^2}$
- (4) $\frac{ML^2}{\pi^2}$

21. A uniform rod of length 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark. A mass of 2 kg is suspended from the rod at 20 cm and another unknown mass ' m ' is suspended from the rod at 160 cm mark as shown in the figure. Find the value of ' m ' such that the rod is in equilibrium:
($g = 10 \text{ m/s}^2$)



- (1) $\frac{1}{2} \text{ kg}$
 - (2) $\frac{1}{3} \text{ kg}$
 - (3) $\frac{1}{6} \text{ kg}$
 - (4) $\frac{1}{12} \text{ kg}$
22. Motion of two balls just before and just after collision are shown in diagram, then velocity $\vec{v}_1$ will be:



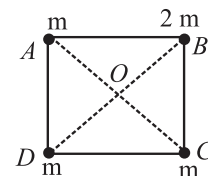
- (1) 7 m/s
 - (2) 5 m/s
 - (3) 12 m/s
 - (4) 10 m/s
23. Two cylinders, one of which is hollow and other solid, have same mass and same moment of inertia about their respective geometrical axis. The ratio of their radii is:
- (1) $1:\sqrt{2}$
 - (2) $1:\sqrt{3}$
 - (3) $\sqrt{3}:\sqrt{3}$
 - (4) 1 : 2

24. A ball of mass m moving with speed 4 m/sec strikes with a ball of triple mass at rest. If coefficient of restitution of collision is 0.5, then velocities of ball after collision will be:

- (1) $\frac{1}{2}, \frac{3}{2}$
- (2) $-\frac{1}{2}, -\frac{3}{2}$
- (3) $-\frac{1}{2}, \frac{3}{2}$
- (4) -1, 3

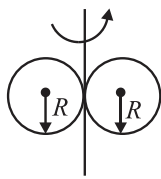
25. The angular velocity ω of a particle varies with time t as $\omega = (10t^2 + 15)\text{rad/s}$. The angular acceleration of the particle at $t = 1\text{s}$ is:
- 10 rad/s^2
 - 20 rad/s^2
 - 5 rad/s^2
 - Zero
26. The angular acceleration of flywheel having moment of inertia 100 kg m^2 , when a torque of 10 Nm is applied on the flywheel is:
- 10 rad/s^2
 - $\pi\text{ rad/s}^2$
 - 0.1 rad/s^2
 - 0.1 $\pi\text{ rad/s}^2$
27. A ball of mass 2 kg is moving with velocity 10 m/s strikes with a ball of mass 3 kg at rest & moves as a combined mass, then their common velocity will be:
- $\frac{10}{3}\text{ m/s}$
 - $\frac{20}{3}\text{ m/s}$
 - 4 m/s
 - 5 m/s
28. Two bodies with masses m_1 and m_2 ($m_1 > m_2$) are joined by a string passing over fixed pulley. Assume masses of the pulley and thread negligible. Then the acceleration of the centre of mass.
- $\left(\frac{m_1 - m_2}{m_1 + m_2}\right)g$
 - $\left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 g$
 - $\frac{m_1 g}{m_1 + m_2}$
 - $\frac{m_2 g}{m_1 + m_2}$
29. A sphere of mass m moving with a constant velocity u hits another stationary sphere of the same mass. If e is the coefficient of restitution, then the ratio of the velocity of two spheres after collision will be:
- $\frac{1-e}{1+e}$
 - $\frac{1+e}{1-e}$
 - $\frac{e+1}{e-1}$
 - $\frac{e-1}{e+1}$

30. When two bodies collide elastically, then:
- kinetic energy of the system alone is conserved.
 - only momentum is conserved.
 - both energy and momentum are conserved.
 - neither energy nor momentum is conserved.
31. A body of mass ' M ' collides against a wall with a velocity v and retraces its path with the same speed. The change in momentum is: (take initial direction of velocity as positive)
- Zero
 - $2Mv$
 - Mv
 - $-2Mv$
32. A gun fires a bullet of mass 50 gm with a speed of 30 m sec^{-1} . Because of this the gun is pushed back with a speed of 1 m sec^{-1} . The mass of the gun is:
- 15 kg
 - 30 kg
 - 1.5 kg
 - 20 kg
33. Centre of mass of the given system of particles will be at:



- OD
 - OC
 - OB
 - AO
34. Two objects of masses 200 gm and 500 gm possess velocities $10\hat{i}\text{ m/s}$ and $(3\hat{i} + 5\hat{j})\text{ m/s}$ respectively. The velocity of their centre of mass in m/s is:
- $5\hat{i} - 25\hat{j}$
 - $\frac{5}{7}\hat{i} - 25\hat{j}$
 - $5\hat{i} + \frac{25}{7}\hat{j}$
 - $25\hat{i} - \frac{5}{7}\hat{j}$

35. Moment of inertia of combination of two discs of same mass M and same radius R kept in contact about the tangent passing through point of contact and in the plane of discs, as shown is:



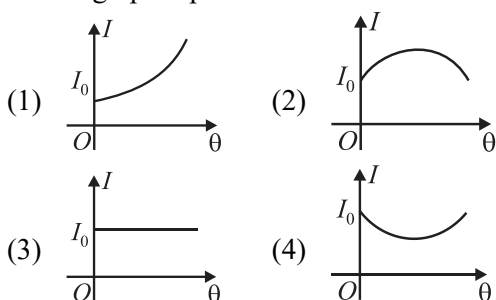
- (1) $\frac{MR^2}{4}$ (2) $\frac{5}{4}MR^2$
 (3) $\frac{MR^2}{2}$ (4) $\frac{5}{2}MR^2$

SECTION - B

36. The moment of inertia of a uniform semicircular wire of mass m and radius r , about an axis passing through its centre of mass and perpendicular to its plane is:

- (1) $\frac{mr^2}{2}$
 (2) mr^2
 (3) $mr^2\left(1 - \frac{4}{\pi^2}\right)$
 (4) $mr^2\left(1 + \frac{4}{\pi^2}\right)$

37. A square plate has a moment of inertia I_0 about an axis lying in its plane, passing through its centre and making an angle θ with one of the sides. Which graph represents the variation of I with θ ?



38. A person of mass m is standing on one end of a plank of mass M and length L and floating in water. The person moves from one end to another and stops. The displacement of the plank is:

- (1) $\frac{Lm}{(m+M)}$
 (2) $Lm(M+m)$
 (3) $L\frac{(m+M)}{m}$
 (4) $\frac{LM}{(m+M)}$

39. Particles of masses $m, 2m, 3m, \dots, nm$ grams are placed on the same line at distance $l, 2l, 3l, \dots, nl$ m from a fixed point. The distance of the centre of mass of the particles from the fixed point in meters is:

- (1) $\frac{(2n+1)l}{3}$
 (2) $\frac{(n+1)l}{2}$
 (3) $\frac{(2n+1)l}{6}$
 (4) $\frac{(n+1)l}{3}$

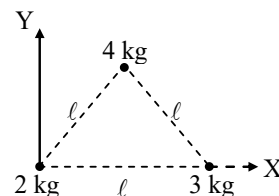
40. A body of mass 5 kg explodes at rest into three fragments with masses in the ratio 1 : 1 : 3. The fragments with equal masses fly in mutually perpendicular directions with speeds of 21 m/s. The speed of the heaviest fragment will be:

- (1) 11.5 m/s
 (2) 14.0 m/s
 (3) 7.0 m/s
 (4) 9.89 m/s

41. Radius of gyration of a rod of Mass M and length l is $l/2$ then distance of axis of rotation from centre of rod is:

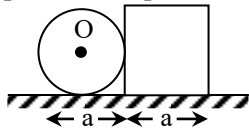
- (1) $l/4$
 (2) $l/6$
 (3) $l\sqrt{6}$
 (4) $l/\sqrt{6}$

42. Three masses 2 kg, 3 kg and 4 kg are lying at the corners of an equilateral triangle of side ℓ . The (X) coordinate of center of mass is:

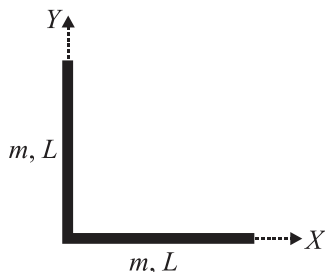


- (1) $\frac{7}{12}\ell$
 (2) $\frac{5}{9}\ell$
 (3) $\frac{7\sqrt{2}}{9}\ell$
 (4) $\frac{\sqrt{5}}{9}\ell$

43. A square plate of edge a and a circular disc of diameter a are placed touching each other at the midpoint of an edge of the plate as shown in figure. Locate the centre of mass of the combination, assuming same mass per unit area for the two plates from point O.



- (1) $\frac{4a}{\pi + 4}$
 (2) $\frac{a}{\pi + 4}$
 (3) $\frac{4}{\pi + 3}$
 (4) None of these
44. Masses 8, 2, 4, 2 kg are placed at the corners A, B, C, D respectively of a square $ABCD$ of diagonal 80 cm. The distance of centre of mass from A will be:
- (1) 20 cm (2) 30 cm
 (3) 40 cm (4) 60 cm
45. Two rods each of mass m and length L are placed along X and Y axis as shown in figure. Coordinates of the centre of mass of the combination is:



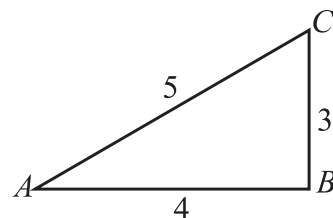
- (1) $\left(\frac{L}{2}, \frac{L}{2}\right)$
 (2) $\left(\frac{L}{4}, \frac{L}{2}\right)$
 (3) $\left(\frac{L}{2}, \frac{L}{4}\right)$
 (4) $\left(\frac{L}{4}, \frac{L}{4}\right)$
46. Two bodies of masses m_1 and m_2 have equal kinetic energies. If p_1 and p_2 are their respective momentum, then ratio $p_1 : p_2$ is equal to:

- (1) $m_1 : m_2$
 (2) $m_2 : m_1$
 (3) $\sqrt{m_1} : \sqrt{m_2}$
 (4) $m_1^2 : m_2^2$

47. A light rod of length l has two masses m_1 and m_2 attached to its two ends. The moment of inertia of the system about an axis perpendicular to the rod and passing through the centre of mass is:

- (1) $\frac{m_1 m_2}{m_1 + m_2} l^2$
 (2) $\frac{m_1 + m_2}{m_1 m_2} l^2$
 (3) $(m_1 + m_2) l^2$
 (4) $\sqrt{m_1 m_2} l^2$

48. The ABC is a triangular plate of uniform thickness. The sides are in the ratio shown in the figure. I_{AB} , I_{BC} and I_{CA} are the moments of inertia of the plate about AB , BC and CA respectively. Which one of the following relations is correct?



- (1) $I_{AB} + I_{BC} = I_{CA}$
 (2) I_{CA} is maximum
 (3) $I_{AB} > I_{BC}$
 (4) $I_{BC} > I_{AB}$
49. What is the torque of the force $\vec{F} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ N acting at the point $\vec{r} = 3\hat{i} + 2\hat{j} + 3\hat{k}$ m about origin?
- (1) $-6\hat{i} + 6\hat{j} - 12\hat{k}$
 (2) $-17\hat{i} + 6\hat{j} + 13\hat{k}$
 (3) $6\hat{i} - 6\hat{j} + 12\hat{k}$
 (4) $17\hat{i} - 6\hat{j} - 13\hat{k}$
50. Find the torque about the origin when a force of $3\hat{j}$ N acts on a particle whose position vector is $2\hat{k}$ m.
- (1) $6\hat{j}$ Nm
 (2) $-6\hat{i}$ Nm
 (3) $6\hat{k}$ Nm
 (4) $6\hat{i}$ Nm

SECTION-(II) CHEMISTRY

SECTION - A

51. For the given exothermic reaction,
 $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) + 2\text{kcal}$,
 the favourable conditions are:
 (1) low temperature, low pressure and catalyst.
 (2) low temperature, high pressure and catalyst.
 (3) high temperature, low pressure and catalyst.
 (4) high temperature, high pressure and catalyst.
52. For the reaction,
 $\text{A}(\text{g}) \rightarrow 2\text{C}(\text{g}) + \text{D}(\text{g})$, the **correct** relation between ΔH and ΔU is:
 (1) $\Delta H = \Delta U$ (2) $\Delta H > \Delta U$
 (3) $\Delta H < \Delta U$ (4) $\Delta H \leq \Delta U$
53. A solution of NaOH contains 0.04 g of NaOH per litre. Its pOH is:
 (1) 10 (2) 9
 (3) 3 (4) 12
54. The following gaseous equilibrium are given
 $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ k_1
 $\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO}$ k_2
 $\text{H}_2 + 1/2\text{O}_2 \rightleftharpoons \text{H}_2\text{O}$ k_3
 The equilibrium constant of the reaction
 $2\text{NH}_3(\text{g}) + 5/2 \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}) + 3\text{H}_2\text{O}(\text{g})$, in
 terms of k_1 , k_2 and k_3 is:
 (1) $\frac{k_1 k_2}{k_3}$ (2) $\frac{k_1 k_3^2}{k_2}$
 (3) $\frac{k_2 k_3^3}{k_1}$ (4) $k_1 k_2 k_3$
55. 15 moles of an ideal gas expands spontaneously into vacuum. The work done by the gas is:
 (1) Zero (2) 10 J
 (3) 20 J (4) 15 J
56. In a vessel containing SO_3 , SO_2 and O_2 at equilibrium, some helium gas is introduced so that the total pressure increases while temperature and volume remains constant. According to Le-Chatelier principle, the dissociation of SO_3 :
 (1) increases.
 (2) decreases.
 (3) remain unaltered.
 (4) change is unpredictable.
57. Which of the following is **incorrect** regarding Gibb's energy?
 (1) It is a state function.
 (2) It is an extensive property.
 (3) Its value is negative for a spontaneous process.
 (4) It is an intensive property.

58. The pK_a of formic acid is 3.75. The pH of a solution containing 0.01 M formic acid and 0.01 M sodium formate is:
 (1) 3.80 (2) 5.00
 (3) 3.75 (4) 4.25
59. An ideal gas is expanded isothermally from 10 L to 20 L at constant pressure of 2.0 atm. What is the change in internal energy of system?
 (1) -30 L-atm (2) +30 L-atm
 (3) -60 L-atm (4) 0 L-atm
60. Match **List-I** (hypothetical reactions) with **List-II** (ratio $\frac{K_p}{K_c}$ for the given reactions) and select the

correct answer using the code given below in the lists:

List-I		List-II	
(A)	$\text{A}_2(\text{g}) + 3\text{B}_2(\text{g}) \rightleftharpoons 2\text{AB}_3(\text{g})$	(I)	$(\text{RT})^{-2}$
(B)	$\text{A}_2(\text{g}) + \text{B}_2(\text{g}) \rightleftharpoons 2\text{AB}(\text{g})$	(II)	$(\text{RT})^0$
(C)	$\text{A}(\text{g}) + 1.5\text{B}_2(\text{g}) \rightleftharpoons \text{AB}_3(\text{g})$	(III)	$(\text{RT})^{\frac{1}{2}}$
(D)	$\text{AB}_2(\text{g}) \rightleftharpoons \text{AB}(\text{g}) + 0.5\text{B}_2(\text{g})$	(IV)	$(\text{RT})^{-\frac{3}{2}}$

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-II, D-IV
 (2) A-III, B-II, C-IV, D-I
 (3) A-II, B-I, C-III, D-IV
 (4) A-I, B-II, C-IV, D-III
61. Among the following, standard enthalpy of formation is zero for:
 (1) C (diamond) (2) $\text{Br}_2(\text{l})$
 (3) $\text{H}_2(\text{l})$ (4) $\text{CO}_2(\text{g})$
62. If the value of an equilibrium constant for a particular reaction is 1.6×10^{12} , then at equilibrium the system will contain:
 (1) mostly reactants.
 (2) mostly products.
 (3) similar amounts of reactants and products.
 (4) all reactants.
63. The reaction, $2\text{A}(\text{s}) + 3\text{B}(\text{g}) \rightarrow \text{A}_2\text{B}_3(\text{s})$ is exothermic in nature. The reaction would be:
 (1) spontaneous at all temperatures.
 (2) spontaneous at low temperature.
 (3) spontaneous at high temperature.
 (4) non spontaneous at all temperatures.

64. The enthalpy of reaction for:
 $2\text{C}(\text{graphite}) + 3\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{C}_2\text{H}_5\text{OH}(\text{l})$ is $-A$ kJ. What is the enthalpy of formation of $\text{C}_2\text{H}_5\text{OH}(\text{l})$?
 (1) $-A \text{ kJ mol}^{-1}$
 (2) $-2A \text{ mol}^{-1}$
 (3) $+A \text{ kJ mol}^{-1}$
 (4) $-4A \text{ kJ mol}^{-1}$
65. The equilibrium constant (K_p) for the reaction $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ is 16. If the volume of the container is reduced to one-half of its original volume, the value of K_p for the reaction at the same temperature will be:
 (1) 32 (2) 64
 (3) 16 (4) 4
66. A system does 200 J of work and at the same time absorbs 100 J of heat. The magnitude of the change in internal energy is:
 (1) 370 J (2) 210 J
 (3) 100 J (4) 60 J
67. At 90°C pure water has $[\text{H}_3\text{O}^+] = 10^{-6} \text{ M}$. What is the value of K_w at this temperature?
 (1) 10^{-6} (2) 10^{-12}
 (3) 10^{-13} (4) 10^{-14}
68. ΔH and ΔS for a reaction are 4500 J mol^{-1} and $4.5 \text{ J K}^{-1} \text{ mol}^{-1}$ respectively. The reaction will be spontaneous:
 (1) above 1000 K.
 (2) at 900 K.
 (3) below 1000 K.
 (4) below 900 K.
69. Consider the reaction,
 $2\text{X}(\text{s}) + \text{Y}(\text{g}) \rightarrow 2\text{Z}(\text{g}) + 700 \text{ kJ}$
 The signs of ΔH , ΔS and ΔG respectively are:
 (1) $\Delta H = -\text{ve}$, $\Delta S = -\text{ve}$, $\Delta G = -\text{ve}$
 (2) $\Delta H = -\text{ve}$, $\Delta S = +\text{ve}$, $\Delta G = +\text{ve}$
 (3) $\Delta H = -\text{ve}$, $\Delta S = -\text{ve}$, $\Delta G = +\text{ve}$
 (4) $\Delta H = -\text{ve}$, $\Delta S = +\text{ve}$, $\Delta G = -\text{ve}$
70. The initial pressure of A is 100 torr. The total pressure of the system becomes 150 torr when the equilibrium, $\text{A}(\text{g}) \rightleftharpoons \text{B}(\text{g}) + \text{C}(\text{g})$ is attained at constant temperature. The value of K_p of this reaction is:
 (1) 15
 (2) 10
 (3) 25
 (4) 50
71. 1.2 moles of an ideal gas expands isothermally and reversibly at 25°C from a volume of 20 L to 40 L. What is the change in entropy of the gas?
 (1) 1.2 R
 (2) $1.2 R \ln 2$
 (3) $2.303 R \ln 2$
 (4) $-R \ln 2$
72. What causes the change in the value of equilibrium constant of any equilibria?
 (1) Adding of inert gas at constant pressure
 (2) Increasing the pressure
 (3) Adding of inert gas at constant volume
 (4) Decreasing the temperature
73. In which of the following reactions entropy increases?
 A. $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
 B. $4\text{Fe}(\text{s}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{Fe}_2\text{O}_3(\text{s})$
 C. $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$
 The **correct** options is/are:
 (1) A only
 (2) B only
 (3) A and B only
 (4) A, B and C only
74. The dissociation constant of monobasic acids A, B and C are 10^{-4} , 10^{-6} and 10^{-10} respectively. The concentration of each monobasic acid is 0.1 M. Which of the following has been arranged in increasing order of pH?
 (1) $\text{C} < \text{B} < \text{A}$
 (2) $\text{A} < \text{B} < \text{C}$
 (3) $\text{B} < \text{C} < \text{A}$
 (4) $\text{B} < \text{A} < \text{C}$
75. If pH of a saturated solution of $\text{Ba}(\text{OH})_2$ is 12, the value of its K_{sp} will be:
 (1) 5.00×10^{-7} (2) 4.00×10^{-6}
 (3) 4.00×10^{-7} (4) 5.00×10^{-6}
76. $\Delta_c H$ of carbon to CO_2 is -400 kJ mol^{-1} then how much heat will be evolved upon formation of 17.6 g CO_2 from carbon and oxygen gas?
 (1) 200 kJ
 (2) 600 kJ
 (3) 540 kJ
 (4) 160 kJ

77. Given below are two statements:
Statement I: The acid-base pair that differs only by one proton is called a conjugate acid-base pair.
Statement II: OH^- is called the conjugate base of an acid H_2O .
 In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
78. If the enthalpy of formation of ammonia is $-A \text{ kJ mol}^{-1}$, the enthalpy change for the reaction,
 $2\text{NH}_3(\text{g}) \rightarrow \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ is:
- (1) $2A \text{ kJ mol}^{-1}$
 - (2) $4A \text{ kJ mol}^{-1}$
 - (3) $8A \text{ kJ mol}^{-1}$
 - (4) $\frac{1}{2} A \text{ kJ mol}^{-1}$
79. A catalyst:
- (1) increases the equilibrium concentration of reactants.
 - (2) decreases the equilibrium concentration of reactants.
 - (3) does not affect equilibrium composition of reactants or products.
 - (4) affects equilibrium composition of reactants or products.
80. The highest pH value is of:
- (1) 0.1 M NaCl
 - (2) $0.1 \text{ M NH}_4\text{Cl}$
 - (3) $0.1 \text{ M CH}_3\text{COONa}$
 - (4) $0.1 \text{ M CH}_3\text{COONH}_4$
81. In which of the following equilibrium equation, $K_p > K_c$?
- (1) $2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$
 - (2) $\text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{PCl}_5(\text{g})$
 - (3) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
 - (4) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
82. In an adiabatic process which of the following is true:
- (1) $q = +w$
 - (2) $-\Delta E = -w$
 - (3) $P\Delta V = 0$
 - (4) $q = \Delta E$

83. $\text{A} + \text{B} \rightleftharpoons \text{C} + 2\text{D}$ equilibrium constant = K_1 ,
 $\text{A} + \text{E} \rightleftharpoons \text{C} + \text{F}$ equilibrium constant = K_2 .
 Then equilibrium constant for the reaction
 $2\text{D} + \text{E} \rightleftharpoons \text{B} + \text{F}$ will be:

- (1) $\frac{K_2}{K_1}$
- (2) $\frac{K_1}{K_2}$
- (3) $K_1 + K_2$
- (4) $K_1 \times K_2$

84. What would happen to a reversible reaction
 $\text{A}_{2(\text{g})} + \text{B}_{2(\text{g})} \rightleftharpoons 2\text{AB}_{(\text{g})}$ at equilibrium when an inert gas is added while the pressure remains unchanged?
- (1) More of the product will be formed.
 - (2) Less of the product will be formed.
 - (3) More of the reactants will be formed.
 - (4) It remains unaffected.
85. The temperature of an ideal gas increases in an:
- (1) adiabatic expansion.
 - (2) isothermal expansion.
 - (3) adiabatic compression.
 - (4) isothermal compression.

SECTION - B

86. In the combustion of 0.4 g of CH_4 , 0.25 kCal of heat is liberated. The heat of combustion of CH_4 is:
- (1) -20 kCal
 - (2) 10 kCal
 - (3) -2.5 kCal
 - (4) -5 kCal
87. The conjugate acid of HPO_3^{2-} is:
- (1) H_3PO_4
 - (2) H_3PO_3
 - (3) H_2PO_3^-
 - (4) PO_4^{3-}
88. The K_a for a weak acid HX is 10^{-6} , then the ionisation constant of X^- will be:
- (1) 10^{-6}
 - (2) 10^{-8}
 - (3) 10^{-14}
 - (4) 10^{-4}

89. The heat of combustion of C_2H_4 , C_2H_6 and H_2 are -1409.5 kJ, -1558.3 kJ and -285.6 kJ. The heat of hydrogenation of ethene is:
- (1) -136.8 kJ
 - (2) -13.68 kJ
 - (3) 273.6 kJ
 - (4) 1.368 kJ
90. For 10^{-3} M CH_3COOH solution if $K_a = 10^{-5}$ then pH is:
- (1) 12
 - (2) 14
 - (3) 4
 - (4) 1
91. Which of the following conditions will always lead to a non-spontaneous change?
- (1) ΔH and ΔS both are positive
 - (2) ΔH is negative and ΔS is positive
 - (3) ΔH and ΔS both are negative
 - (4) ΔH is positive and ΔS is negative
92. The pH of a 0.02 M ammonia solution which is 5% ionised will be:
- (1) 2
 - (2) 11
 - (3) 5
 - (4) 7
93. Heat evolved in the reaction $H_2 + Cl_2 \rightarrow 2HCl$ is -182 kJ. Bond energies of $H-H$ and $Cl-Cl$ are 430 and 242 kJ mol^{-1} respectively. The $H-Cl$ bond energy is:
- (1) 245 kJ mol^{-1}
 - (2) 427 kJ mol^{-1}
 - (3) 336 kJ mol^{-1}
 - (4) 154 kJ mol^{-1}
94. The work done by a system is 8 J when 40 J heat is supplied to it. The change in internal energy of the system during the process:
- (1) 32 J
 - (2) 40 J
 - (3) 48 J
 - (4) -32 J
95. For the given reaction:
 $CO_2(g) + H_2(g) \rightarrow CO(g) + H_2O(g)$; $\Delta H = 40$ kJ
 The ΔH is specifically called:
- (1) heat of formation of CO .
 - (2) heat of combustion.
 - (3) heat of reaction.
 - (4) heat of hydrogenation of $C = O$ bond.
96. If 100 mL of pH = 3 and 400 mL of pH = 3 is mixed, what will be the pH of the mixture?
- (1) 3.2
 - (2) 3.0
 - (3) 3.5
 - (4) 2.8
97. Given below are two statements:
Statement I: HCl acid will apply common ion effect on the ionization of HCN .
Statement II: Degree of ionization of HCN is increased in presence of HCl .
 In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
98. Evaluate K_p for the reaction: $H_2 + I_2 \rightleftharpoons 2HI$. If 2 moles each of H_2 and I_2 are taken initially. The equilibrium moles of HI are 2.
- (1) 2.5
 - (2) 4
 - (3) 0.25
 - (4) 1.0
99. If K_a of HCN is 4×10^{-10} having concentration 2.5×10^{-1} M, then pH of HCN (aq) is:
- (1) 4.2
 - (2) 4.7
 - (3) 0.47
 - (4) 5.0
100. The solubility product of CuS , Ag_2S and HgS are 10^{-37} , 10^{-44} and 10^{-54} respectively. The solubility of these sulphides will be in the order:
- (1) $HgS > Ag_2S > CuS$
 - (2) $Ag_2S > HgS > CuS$
 - (3) $CuS > Ag_2S > HgS$
 - (4) $Ag_2S > CuS > HgS$

SECTION-(III) BOTANY

SECTION - A

101. Order Polymoniales includes the families.

- A. Poaceae
- B. Solanaceae
- C. Convolvulaceae
- D. Anacardiaceae

Choose the most appropriate answer from the options given below:

- (1) A and B only
- (2) A and C only
- (3) B and D only
- (4) B and C only

102. The number and different types of organisms present on earth is refers to:

- (1) population.
- (2) variety.
- (3) biodiversity.
- (4) taxa.

103. Identify the **incorrect** statement for binomial nomenclature system.

- (1) Biological names are either derived from Latin language or Latinised.
- (2) Biological names are printed in italics.
- (3) Name of the author is written at the start of the biological name.
- (4) Each organism is given only one biological name consisting of two parts.

104. Systematics:

- (1) is derived from a greek word.
- (2) does not include classification of organisms.
- (3) considers taxonomy along with evolutionary relationships.
- (4) does not deal with diversity of organisms.

105. The number of species that are known and described ranges between:

- (1) 7 - 8 million.
- (2) 1.7 - 1.8 million.
- (3) 2 - 3 million.
- (4) 2.7 - 3.8 million.

106. The families like Felidae and Canidae are included in the same taxonomic category named:

- (1) carnivora.
- (2) polymoniales.
- (3) primata.
- (4) diptera.

107. How many statement(s) is/are **correct**?

- A. Reproduction is not defining property because it occurs in sterile human couple.
- B. Reproduction is defining characteristics of living organism.
- C. Self-consciousness is seen only in human.
- D. Consciousness is defining property of living organism.
- E. Metabolism is defining property of all living organism without exception.

- (1) One
- (2) Two
- (3) Three
- (4) Five

108. Choose the **incorrect** statement from the following.

- (1) The genus *Panthera* differs from another genus *Felis* which includes cats.
- (2) Monkey, gorilla and gibbon are placed under a common group called Mammalia.
- (3) Tiger, cat and dog are place in a common group called Mammalia.
- (4) Insecta is placed under phylum Chordata.

109. In case of plants, classes with a few similar characters are assigned to a higher category called _____.

- (1) Class
- (2) Division
- (3) Order
- (4) Family

110. Species *sapiens* belongs to which of the given higher taxonomic categories?

- (1) Anacardiaceae.
- (2) Muscidae.
- (3) Chordata.
- (4) Poaceae.

111. The scientific names for plants and animals are provided by _____ and _____ respectively.

- (1) ICZN, ICBN
- (2) ICNB, ICZN
- (3) ICBN, ICZN
- (4) ICNZ, ICNB

112. Read the following and identify the number of **correct** statement(s) related to biological classification of Mango.

- A. It belong to the genus *Mangifera*.
- B. It belong to the class Angiospermae.
- C. It belong to the order Anacardiaceae.
- D. It belong to the division Monocotyledonae.

- (1) One
- (2) Two
- (3) Three
- (4) Four

113. Considering the hierarchical arrangement of taxonomies, order lies between:

- (1) genus and species.
- (2) genus and family.
- (3) family and class.
- (4) class and phylum.

114. All the following species belong to same genus, **except**:

- (1) *tuberosum*.
- (2) *aestivum*.
- (3) *melongena*.
- (4) *nigrum*.

115. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: There is a need to standardise the naming of living organisms such that a particular organism is known by the same name all over the world.

Reason R: The local names of living organisms vary from place to place even within a country.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

116. In which organisms, reproduction is synonymous with growth?

- (1) Bacteria
- (2) Unicellular algae
- (3) *Amoeba*
- (4) All of these

117. Naming is only possible when the organism is described correctly and we know to what organism the name is attached to. This is:

- (1) characterisation.
- (2) identification.
- (3) nomenclature.
- (4) classification.

118. How many statement(s) is/are **correct**?

- A. Growth is reversible in living and irreversible in non living.
 - B. Growth is defining property of organism.
 - C. Cellular organisation of the body is the defining feature of life forms.
 - D. Growth and reproduction is synonym in multicellular organism.
 - E. Growth and reproduction is exclusive property in unicellular organism.
- (1) One
 - (2) Two
 - (3) Three
 - (4) Five

119. As we go from kingdom to species in a taxonomic hierarchy, the number of common characteristics:

- (1) goes on decreasing.
- (2) may increase or decrease.
- (3) remains same.
- (4) goes on increasing.

120. Read the following and choose the set of **correct** statements.

- A. All taxonomic categories together constitute the taxonomic hierarchy.
- B. Each taxonomic category represents a rank and is commonly termed as taxon.
- C. Biologist follow universally accepted principles to provide scientific names to known organisms.
- D. All organisms, including those in the plant and animal kingdoms have species as the highest category.

Choose the most appropriate answer from the options given below:

- (1) A, C and D only
- (2) A, B and C only
- (3) A, B and D only
- (4) A, B, C and D

121. Identify the **odd** one out w.r.t. *Panthera leo*.

- (1) Common name of leopard.
- (2) *Panthera* represents generic name.
- (3) *leo* represents specific epithet.
- (4) *Panthera* represents higher level of taxon than *leo*.

122. Match List-I with List-II:

List-I		List-II	
(A)	<i>Triticum aestivum</i>	(I)	Housefly
(B)	<i>Solanum melongena</i>	(II)	Tiger
(C)	<i>Musca domestica</i>	(III)	Wheat
(D)	<i>Panthera tigris</i>	(IV)	Brinjal

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-I, C-III, D-II
- (3) A-III, B-I, C-II, D-IV
- (4) A-IV, B-III, C-I, D-II

123. Read the following and choose the set of correct statements.

- A. To form a taxonomic category, the basic requirement is the knowledge of characters of an individual or group of organisms.
- B. The living world has variety of organism.
- C. Classification is the process by which anything is grouped into convenient categories.
- D. Binomial nomenclature is given by Carolus Linnaeus.

Choose the most appropriate answer from the options given below:

- (1) A, C and D only
- (2) A, B and C only
- (3) A, B and D only
- (4) A, B, C and D

124. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The taxonomic groups are distinct biological entities and not merely morphological aggregates.

Reason R: Taxonomical studies of all known organisms have led to the development of categories.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

125. Based on common features like presence of notochord and dorsal hollow neural system organisms are included in phylum:

- (1) Solanaceae.
- (2) Mammalia.
- (3) Chordata.
- (4) Arthropoda.

126. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Lower the taxa, more are the characteristics that the members within the taxon share.

Reason R: Higher the category, lesser is the difficulty of determining the relationship to other taxa at the same level.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

127. Match List-I with List-II:

List-I		List-II	
(A)	Family	(I)	Diptera
(B)	Order	(II)	Arthropoda
(C)	Class	(III)	Hominidae
(D)	Phylum	(IV)	Monocotyledonae

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-IV, B-II, C-I, D-III
- (3) A-III, B-I, C-IV, D-II
- (4) A-III, B-II, C-IV, D-I

128. Read the following statements.

- A. Naming of an organism is not possible without identification.
- B. Each rank or taxon, represents a unit of classification.
- C. Genera are aggregates of closely related species.
- D. Family has a group of related genera with a greater number of similarities as compared to genus and species.
- E. Carnivora includes tiger, cat, dog.

How many is/are **correct**?

- (1) One
- (2) Two
- (3) Four
- (4) Five

129. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Families are characterised on basis of both vegetative and reproductive features of plants species.

Reason R: The categories like species, genus and families are based on a number of similar characters.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

130. The scientific nomenclature representing the species of wheat is:

- (1) *Triticum*.
- (2) *Aestivum*.
- (3) *aestivum*.
- (4) *Triticum aestivum*.

131. Basis of taxonomy are:

- (1) characterisation and classification only.
- (2) characterisation, identification and classification only.
- (3) characterisation, classification and nomenclature only.
- (4) characterisation, identification, classification and nomenclature.

132. How many of the following taxa represents hierarchy above family?

Canidae, Polymoniales, Diptera, Poales, Anacardiaceae

- (1) Two
- (2) Three
- (3) Four
- (4) Five

133. Find the **incorrect** one.

- (1) An organism represents/occupies a place or position in the system of classification.
- (2) The local names of plants and animals vary from place to place, even within a country.
- (3) If we were to visit a dense forest, we would probably see a much greater number and kinds of living organisms in it.
- (4) Biological names can be written in any language.

134. Lion, leopard and tiger are species of the genus:

- (1) *Felis*.
- (2) *Panthera*.
- (3) *Canis*.
- (4) *Musca*.

135. Which of the following is **not** a component of binomial name?

- (1) Family name
- (2) Generic name
- (3) Specific epithet
- (4) Name of author

SECTION - B

136. The Muscidae is equivalent to _____ of mango.

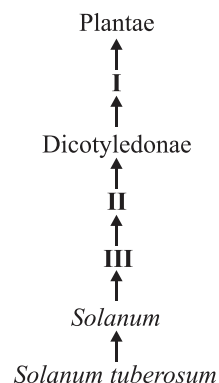
Complete the statement by selecting **correct** option to fill up the blank.

- (1) Sapindales
- (2) Anacardiaceae
- (3) Dicotyledonae
- (4) Poales

137. Scientific name of man in *Homo sapiens* Linn in the above name Linn refers to:

- (1) genus of man.
- (2) species of man.
- (3) a scientist who changed the name proposed by Linnaeus and proposed present name.
- (4) a scientist who for the first time described man.

138. Identify **I**, **II** and **III** and choose **correct** answer:



	I	II	III
(1)	Angiospermae	Sapindales	Solanaceae
(2)	Angiospermae	Polymoniales	Solanaceae
(3)	Angiospermae	Polymoniales	Anacardiaceae
(4)	Angiospermae	Sapindales	Anacardiaceae

139. Three genera *Datura*, *Petunia* and *Solanum* share common vegetative and reproductive characteristics. On this basis they are placed in same:
- (1) order. (2) family.
(3) class. (4) division.

140. Out of the following taxa which one represent the highest rank in taxonomic hierarchy?
- (1) Poales
(2) Anacardiaceae
(3) Monocotyledonae
(4) Polymoniales

141. Match List-I with List-II:

List-I (Common Name)		List-II (Taxonomic category order)	
(A)	Wheat	(I)	Primata
(B)	Lion	(II)	Diptera
(C)	Housefly	(III)	Carnivora
(D)	Man	(IV)	Poales

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-IV, D-III
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-IV, C-I, D-III
(4) A-III, B-IV, C-II, D-I
142. Genus *Solanum* includes:
- A. potato. B. brinjal.
C. *Datura*. D. *Petunia*.
(1) A and C only (2) A and B only
(3) A and D only (4) A, B, C and D
143. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Species is a group of individuals with fundamental similarities.
Reason R: *indica*, *leo*, *tuberosum* represent such group of individuals.
In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

144. Gorilla and monkey belong to same:

- (1) species. (2) genus.
(3) division. (4) order.

145. Identify the **incorrect** statement(s).

- A. Biological names are not assigned for bacteria.
B. Category denotes rank.
C. Insects represent a group of organisms sharing common features like four pairs of jointed legs.
D. Each genus may have one or more than one specific epithets representing different organisms.

Choose the most appropriate answer from the options given below:

- (1) A only
(2) B, C and D only
(3) A and C only
(4) A and D only

146. All plants are included in:

- (1) Division - Angiospermae
(2) Class - Monocotyledonae
(3) Family - Poaceae
(4) Kingdom - Plantae

147. Which one of the following is **not** a category?

- (1) Class
(2) Animalia
(3) Phylum
(4) Kingdom

148. Select the **correct** option w.r.t. scientific name of man when it is handwritten.

- (1) *Homo sapiens*
(2) Homo Sapiens
(3) Homo sapiens
(4) Homo sapiens

149. Which of the following represents a Phylum?

- (1) Chordata
(2) Carnivora
(3) Angiospermae
(4) All of these

150. In taxonomic hierarchy of animals, classes with a few similar characters are assigned to a higher category called:

- (1) Phylum.
(2) Order.
(3) Family.
(4) Division.

SECTION-(IV) ZOOLOGY

SECTION - A

- 151.** Inactive fibrinogen are converted into active fibrin in the plasma by the enzyme
(1) carbonic anhydrase.
(2) thrombokininase.
(3) thrombin.
(4) succinic dehydrogenase.
- 152.** The ventricular contraction starts shortly:
(1) after Q wave.
(2) before Q wave.
(3) after R wave.
(4) before R wave.
- 153.** A special coronary system of blood vessels is present in our body exclusively for the circulation of blood to and from the:
(1) pulmonary arteries.
(2) cardiac musculature.
(3) hepatic vein.
(4) wall of blood vessels.
- 154.** All of the following together make about 6-8 per cent of blood plasma **except**;
(1) Albumin. (2) Globulin.
(3) Fibrinogen. (4) Water.
- 155.** Erythroblastosis foetalis can be avoided by administering X to the mother immediately after the delivery of the Y
Choose the option which fills the blank **correctly**.
(1) X-anti-Rh antibodies; Y-first child.
(2) X-anti-Rh antigen; Y-first child.
(3) X-anti-B antibodies; Y-second child.
(4) X-anti-A antibodies; Y- second child.
- 156.** During cardiac cycle the flow of blood into the ventricles increases by about 30 per cent during:
(1) atrial systole. (2) ventricular systole.
(3) joint diastole. (4) atrial diastole.
- 157.** Given below are two statements:
Statement I: Lymphatic system collects the lymph and drains it back to the major veins.
Statement II: Open circulatory system is considered to be more advantageous.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.
- 158.** Choose the **odd** one out w.r.t. four chambered heart.
(1) Crocodiles (2) Birds
(3) Mammals (4) Fish
- 159.** All the following statements are true for lymph **except**:
(1) It has the same mineral distribution as that in plasma.
(2) Exchange of nutrients and gases between the blood and the cells always occur through this fluid.
(3) It is a colourless fluid containing specialised lymphocytes which provide immunity.
(4) Fats are absorbed through lymph in the lacteals present in the stomach villi.
- 160.** Match **List-I** and **List-II**.
- | List-I | | List-II | |
|--------|--------------------|---------|----------------------------------|
| (A) | Leucocytes | (I) | 6-8 percent of whole leucocyte |
| (B) | Average heart beat | (II) | 6000-8000 mm ⁻³ |
| (C) | Neutrophils | (III) | 60-65 percent of whole leucocyte |
| (D) | Monocytes | (IV) | 72 times/minute |
- Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-II, D-I
(2) A-I, B-IV, C-II, D-III
(3) A-IV, B-II, C-I, D-III
(4) A-II, B-IV, C-III, D-I
- 161.** Which of the following feature is common in both fish and humans?
(1) Presence of pulmonary circulation.
(2) Numbers of chambers of heart.
(3) Heart pumps only deoxygenated blood.
(4) Presence of closed circulatory system.
- 162.** The **correct** route of origin and conduction of impulse in the heart can be represented by:
(1) AV node → Purkinje fibre → Bundle of His → SA node
(2) Purkinje fibre → AV node → Bundle of His → SA node
(3) AV node → SA node → Purkinje fibre → Bundle of His
(4) SA node → AV node → Bundle of His → Purkinje fibre

163. Match List-I and List-II.

List-I		List-II	
(A)	Clenched fist	(I)	Deoxygenated blood
(B)	Pulmonary artery	(II)	Oxygenated blood
(C)	Aorta	(III)	Essential for blood clotting.
(D)	Calcium ions	(IV)	Size of heart

Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-III, D-II
- (2) A-I, B-IV, C-II, D-III
- (3) A-IV, B-I, C-II, D-III
- (4) A-II, B-IV, C-III, D-I

164. Reduced count of which of the following formed elements leads to clotting disorders?

- (1) Lymphocytes. (2) Thrombocytes.
- (3) Monocytes. (4) Erythrocytes.

165. The blood from the intestine is carried to the liver by ___X___ prior to its return to the heart.

Choose the option which fills the blank **correctly**.

- (1) hepatic artery.
- (2) hepatic portal vein.
- (3) renal vein.
- (4) renal artery.

166. During which phase of cardiac cycle maximum filling of ventricle takes place?

- (1) Ventricular systole.
- (2) Joint diastole.
- (3) Atrial systole.
- (4) Ventricular diastole.

167. Select the **incorrect statement w.r.t vein.**

- (1) The wall of vein consists of three layers.
- (2) Tunica intima is made up of squamous endothelium.
- (3) Tunica media is made up of smooth muscle and elastic fibres.
- (4) The tunica media is comparatively thick in the veins than arteries.

168. Choose the **incorrect option w.r.t angina pectoris.**

- (1) It occurs when no oxygen reach to the heart muscle.
- (2) It can occur in man and woman of any age.
- (3) It is less common in elderly.
- (4) It occurs due to condition that effect blood flow.

169. Protein makes 6-8 percent of plasma which is less to per cent of ___X___ of total leucocyte.

Choose the option which fills the blank **correctly**.

- (1) eosinophils
- (2) basophils
- (3) monocytes
- (4) lymphocytes

170. In an individual left ventricle pumps out approximately 80 mL of blood during each cardiac cycle with heartbeat 75 beats per minute then calculate the cardiac output.

- (1) 6000 mL
- (2) 600 mL
- (3) 5000 mL
- (4) 5040 mL

171. Read the following statements (A-D) and select the **incorrect statements.**

- A. A special neural centre in the medulla oblongata can moderate the cardiac function.
- B. Neural signals through the sympathetic nerves can decrease the rate of heart beat.
- C. Parasympathetic neural signals increase the rate of heart beat.
- D. Heart is situated in the thoracic cavity in between the two lungs.

Choose the most appropriate answer from the options given below:

- (1) A, B, and D only
- (2) B, C, and D only
- (3) B and C only
- (4) A, B, C and D

172. In human heart, the atrium and ventricle of same side are separated by;

- (1) thin fibrous atrio-ventricular septum.
- (2) thick muscular atrio-ventricular septum.
- (3) thick fibrous atrio-ventricular septum.
- (4) thin muscular atrio-ventricular septum.

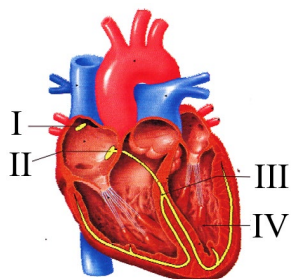
173. Read the following statements (A-E) and select the **incorrect statement.**

- A. RBCs are devoid of nucleus in most of the mammals and are biconvex in shape.
- B. A healthy individual has 70-75 gms of haemoglobin in every 100 ml of blood.
- C. The colour of RBCs is red due to presence of haemoglobin.
- D. Annelids have closed circulating system.
- E. Granulocyte include lymphocyte, neutrophils and basophils.

Choose the most appropriate answer from the options given below:

- (1) A, B, D and E only
- (2) B, C, D and E only
- (3) A, B and C only
- (4) A, B and E only

174. The valve which guards the opening between the right atrium and the right ventricle is made up of:
- (1) three fibrous flaps.
 - (2) three muscular flaps.
 - (3) two muscular flaps.
 - (4) two fibrous flaps.
175. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Human heart is myogenic.
Reason R: Normal activities of the heart are regulated intrinsically by specialised nodal tissue.
 In the light of the above statements, choose the correct answer from the options given below:
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is not the correct explanation of A.
176. Which of the following is **correct** about complete double circulation?
- (1) There are two separate circulatory pathways.
 - (2) There is no mixing of oxygenated and deoxygenated blood.
 - (3) The left atrium receive oxygenated blood and the right atrium receive deoxygenated blood.
 - (4) All of these
177. The diagram given below shows a section of a human heart with labelled I, II, III and IV. Select the **correct** option which gives **correct** identification.



- (1) I-Sino-atrial node present in the right lower corner of the left atrium.
- (2) II-Atrio-ventricular node present in the upper left corner of the left atrium.
- (3) III-Bundle of His transmits action potential to the ventricles.
- (4) IV-Thin -walled, the inter-ventricular septum, separates the left and the right ventricles.

178. An individual with anti -B antibodies in his plasma can donate blood to all of the following **except**:
- (1) A person with no antibodies against blood antigen in his plasma.
 - (2) A person with both A and B antigen on his plasma RBCs.
 - (3) A person with A antigen on his RBCs.
 - (4) A person with both anti-A and anti-B antibodies in his plasma.
179. Which of the following does **not** contain clotting factors?
- (1) Serum
 - (2) Plasma
 - (3) Lymph
 - (4) Blood
180. Which of the following is the leucocyte that destroy foreign organisms entering the body?
- (1) Eosinophils
 - (2) Monocytes
 - (3) Basophils
 - (4) Thrombocytes
181. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In amphibians and reptiles the left atrium receives oxygenated blood and right atrium receives deoxygenated blood.
Reason R: The oxygenated blood get mixed up with deoxygenated blood in double ventricle in amphibians and reptiles.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is not the correct explanation of A.
182. Humans have four type of blood group due to;
- (1) specific type of haemoglobin in RBCs.
 - (2) specific type of antigens on the surface of RBCs.
 - (3) specific type of antigens on the surface of WBCs.
 - (4) specific type of antibodies on the surface of RBCs.
183. Purkinje fibres arise from:
- (1) middle of ventricle.
 - (2) anterior part of atria.
 - (3) apex of ventricle.
 - (4) dorsal part of atria.
184. Which of the following option is **correct** about first heart sound?
- (1) Lub during about the closure of atrioventricular valve.
 - (2) Dub during the closure of semilunar valve.
 - (3) Dub during the closure of atrioventricular valve.
 - (4) Lub during the closure of semilunar valve.
185. How many double circulation are normally completed by human heart in one minute?
- (1) 15
 - (2) 60
 - (3) 72
 - (4) 8

SECTION - B

186. Choose the **odd** one out w.r.t the presence of nucleus.

- (1) Erythrocytes. (2) Lymphocytes.
(3) Monocytes. (4) Neutrophils.

187. Identify the given below diagram and select the **correct** option.



(1)	Eosinophils	resist infections and are also associated with allergic reactions.
(2)	Thrombocytes	phagocytic in nature.
(3)	Monocytes	helps in clotting of blood.
(4)	RBCs	Secrete histamine, serotonin and heparin.

188. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Heart failure is sometimes called congestive heart failure.

Reason R: Congestion of the lungs is one of the main symptoms of heart failure.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

189. Semilunar valve closes when;

- (1) atrial pressure rises.
(2) atrial pressure fall.
(3) ventricular pressure falls.
(4) ventricular pressure rises.

190. Read the following statements (A-E) and select the **incorrect** option.

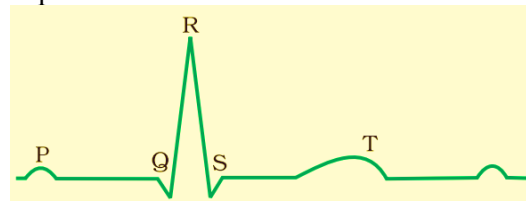
- A. The Rh antigen is present in nearly 80 per cent of humans on the surface of WBCs.
B. ABO group should also be matched before blood transfusions.
C. Electro-cardiograph is the graphical representation of the electrical activity of the heart during a cardiac cycle.
D. Lymphocytes is a granulocyte WBCs.
E. Systemic circulation provides nutrients, O_2 and other essential substances to the tissues.

Choose the most appropriate answer from the options given below:

- (1) A, B, D and E only
(2) B, C, D and E only
(3) A, B, C and E only
(4) A, C and D only

191. The given below shows a diagrammatic presentation of a standard ECG.

Select the **correct** option which gives correct explanation.



- (1) By counting the number of P wave one can determine the heart beat rate of an individual.
(2) The T-wave represents the repolarisation of the auricles.
(3) P-wave represents the depolarisation of the ventricle.
(4) Any deviation in the shape of standard ECG indicates a possible abnormality.

192. Normal blood pressure of a healthy individual is X .

Choose the option which fills the blank **correctly**.

- (1) 140/90 mm Hg (2) 120/80 mm Hg
(3) 80/120 mm Hg (4) 90/40 mm Hg

193. Given below are two statements.

Statement I: The human body has the ability to alter the stroke volume as well as the heart rate and thereby the cardiac output.

Statement II: The cardiac output of an athlete will be much higher than that of an ordinary man.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

194. Match **List-I** and **List-II**.

List-I		List-II	
(A)	Hypertension	(I)	Acute chest pain
(B)	Atherosclerosis	(II)	Heart stops beating
(C)	Angina pectoris	(III)	Affects vital organs
(D)	Cardiac arrest	(IV)	Lumen of arteries become narrower

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
(2) A-I, B-IV, C-II, D-III
(3) A-III, B-I, C-IV, D-II
(4) A-II, B-IV, C-III, D-I

195. The total duration of cardiac cycle is:

- (1) 0.7 seconds. (2) 0.8 seconds
(3) 0.5 seconds. (4) 0.4 seconds.

196. Different types of heart and blood circulation pattern are seen in different animals. Which of the following is **incorrect** w.r.t animal, type of heart and type of circulation?

- (1) Fishes have a 2-chambered heart with an atrium and a ventricle and exhibit single circulation.
- (2) Crocodiles have a 3-chambered and exhibit incomplete double circulation.
- (3) Birds have 4-chambered heart and exhibit complete double circulation.
- (4) Mammals have two atria and two ventricles and they exhibit complete double circulation.

197. Complete the analogy w.r.t normal count present in blood per mm^3

Thrombocytes: 1, 500, 00-3, 500, 00 :: Erythrocytes : ____.

Choose the option that fills the blank **correctly**.

- (1) 5.5 billions
- (2) 5.5 millions
- (3) 4.5 billions
- (4) 4 millions

198. Given below are two statements:

Statement I: Chordae tendinae is present in only left ventricle of human heart.

Statement II: Mitral valve guards the opening between the left atrium and the left ventricle.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

199. To obtain a standard ECG a patient is connected to the machine with;

- (1) three electrical leads one to each wrist and to the right ankle.
- (2) two electrical leads one to ankle and to the left wrist.
- (3) three electrical leads one to each wrist and to the left ankle.
- (4) two electrical leads one to left wrist and to the right ankle.

200. Blood protein which primarily helps in maintaining the osmotic balance is:

- (1) globulins. (2) albumin.
- (3) fibrinogen. (4) prothrombin.

